

Notes: Henderson, Squires, Storeygard, and Weil (QJE, 2018)

The Global Distribution of Economic Activity: Nature, History, and the Role of Trade

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1 Big picture

- **Question.** How much of the modern spatial distribution of economic activity can be explained by first-nature geography, and why do agricultural fundamentals matter more in today's early-developing countries while trade fundamentals matter more in today's late-developing countries?
- **Setting.** The paper studies the global distribution of lights at night across roughly **242,184 quarter-degree land grid cells**. It combines these lights data with **24 physical geography attributes** and with country-level measures of historical development timing.
- **Core puzzle.** Among countries that developed early, agricultural variables explain far more within-country variation in lights than trade variables. Among countries that developed late, the ratio is much smaller even though these countries are more agricultural today.
- **Main conceptual contribution.** The paper combines *first nature* with *history and persistence*. Geography matters, but the weight attached to different geographic features depends on when a country urbanized relative to two historical changes: rising agricultural productivity and falling transport costs.
- **Main mechanism.**
 - In early developers, structural transformation started while transport costs were still high, so cities formed in multiple agriculturally productive regions.
 - In late developers, transport costs fell before structural transformation, so manufacturing concentrated in relatively few, often coastal, locations.
 - Because cities persist, the modern spatial pattern still reflects the geography that mattered at the moment of agglomeration.
- **Main empirical design.** The authors regress log lights on detailed geography, first globally and then with country fixed effects. They then allow the coefficients on geography to differ between early and late agglomerators using interactions and data-driven country splits.
- **Main baseline result.** Geography alone explains about **47%** of worldwide variation in lights and about **35%** of within-country variation. In the fixed-effects regression, the overall R^2 is **0.577**; country fixed effects alone explain **0.345**, so geography contributes **0.232** more, which is about 35% of the remaining within-country variation.
- **Key heterogeneity result.** In early developers, agricultural variables incrementally explain at least **six times** as much variation in lights as trade variables; in late developers the ratio is only about **1.5**. The double differential is about **0.17** no matter whether countries are split by schooling, urbanization, or GDP per capita in 1950.
- **Key marginal-effect result.** As a group, agricultural variables have **19–33% larger** absolute marginal effects in early developers, while trade variables have **39–65% smaller** absolute marginal effects.
- **Key spatial-inequality result.** Countries that developed earlier are more spatially equal today. They have lower spatial Ginis for both lights and educational attainment.

- **Main credibility result.** The central pattern is not just an Old World versus New World difference, not just a variance comparison, and not just a cross-country institutions story. It remains visible within countries, with country fixed effects and with multiple alternative split measures.
- **Economic takeaway.** Poor and middle-income countries should not be expected to replicate the interior-city pattern of countries that urbanized much earlier. Modern geography matters through the filter of when agglomeration happened.

2 Data and measurement (historical context and key objects)

2.1 Historical background: why agriculture, transport, and timing matter

- The historical motivation rests on two long-run changes:
 - rising labor productivity in agriculture, which released workers from farming and allowed urbanization;
 - falling transport costs, which made it less necessary for cities to sit near local food-producing areas and made coastal and trade-access locations more valuable.
- In early developers, urbanization began before the dramatic fall in transport costs. In late developers, transport costs had already fallen substantially before large-scale structural transformation began.
- The paper also relies on urban persistence: once cities form, they are hard to move.

Interpretation. The modern map of economic activity is informative about history because old agglomerations persist even when technology changes.

2.2 Outcome variable: lights at night and the quarter-degree grid

- The main dependent variable is radiance-calibrated nighttime lights for **2010**.
- The raw lights pixels are about 1/120 of a degree; the analysis aggregates everything to **quarter-degree cells**.
- The sample contains **242,184 land cells** worldwide.
- The cell-level outcome is

$$\ln(\text{light}_{ic}) = \ln \left(\max \left\{ \frac{\sum_{j \in i} \text{light}_{jc} 1(\text{land}_{jc})}{\sum_{j \in i} 1(\text{land}_{jc})}, 0.0034 \right\} \right). \quad (1)$$

Interpretation.

- Dividing by the number of land pixels removes distortions from cells that are partly water or ice.
- The bottom code of **0.0034** treats observed zeroes as censoring rather than literal absence of light.
- Within countries, lights mainly proxy for the population distribution rather than pure income-per-capita variation.

2.3 Base geography variables

The paper separates geography into **base**, **agriculture**, and **trade** blocks.

- The two base variables are:
 - **ruggedness**, which raises both agricultural and transport costs;
 - **malaria stability**, which affects the cost of human settlement more generally.

Interpretation. These are not cleanly “agriculture” or “trade” variables, so the paper treats them separately.

2.4 Agriculture variables

The agriculture block includes:

- temperature,
- precipitation,
- growing days,
- land suitability for cultivation,
- elevation,
- absolute latitude,
- and a set of biome indicators (tropical forest, temperate forest, grassland, tundra, Mediterranean, mangroves, desert, and related categories).

Interpretation. These variables proxy for the local productivity of land in supplying food and supporting dense settlement before cheap transport and modern structural transformation.

2.5 Trade variables

The trade block focuses on access to water transport.

- coast (indicator),
- distance to coast,

- natural harbor within 25 km,
- navigable river within 25 km,
- major lake within 25 km.

Interpretation. These variables capture first-nature advantages for moving goods domestically and internationally.

2.6 Early- versus late-agglomerator measures

The paper uses three country-level indicators of whether agglomeration happened early:

- average years of schooling in **1950**,
- urbanization rate in **1950**,
- GDP per capita in **1950**.

The data-driven cutoffs that minimize the summed residual variance are:

- **2.83** years of schooling,
- **36.16%** urbanized,
- **\$2,231** GDP per capita (2005 PPP dollars).

Interpretation. Countries above these thresholds are treated as having transformed and agglomerated relatively early.

2.7 Spatial inequality outcomes

Beyond lights, the paper studies the spatial distribution of human capital.

- A **spatial Gini of lights** is computed across quarter-degree cells within each country.
- A **regional education Gini** is computed from first-level administrative regions using the subnational schooling data of Gennaioli et al.

Interpretation. If late developers agglomerate into fewer coastal centers, they should be more spatially unequal in both output and skills.

2.8 Key descriptive facts

- Mean $\ln(\text{light}/\text{land pixels})$ is **-3.357** with standard deviation **3.119**.
- Mean malaria index is **1.921**; mean ruggedness is **2.781**.
- Mean land suitability is **0.275**; mean growing days are **139.6**.
- Mean distance to coast is **0.486 thousand km**; **9.72%** of cells are coastal.

- The baseline global regression without country fixed effects has $R^2 = 0.467$; with country fixed effects it has $R^2 = 0.577$.

Interpretation. A surprisingly parsimonious geography vector accounts for a large share of the global map of economic activity.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework: two regions, two sectors, and persistence

The model in the paper’s online appendix has:

- two regions: **coast** and **hinterland**;
- two sectors: **food** and **manufacturing**;
- agglomeration economies and congestion in urban manufacturing;
- costly trade in goods across regions.

The timing of two shocks matters:

- an increase in agricultural productivity,
- a fall in transport costs.

Interpretation.

- If agricultural productivity rises first while transport is still expensive, cities form in multiple interior agricultural regions.
- If transport costs fall first, manufacturing can concentrate in one or a few coastal places and then persist there.

3.2 Baseline geography regression

The baseline estimating equation is

$$\ln(\text{light}_{ic}) = X_{ic}\beta + \varepsilon_{ic}, \quad (2)$$

where X_{ic} contains the full set of geography controls.

The paper estimates this equation:

- without country fixed effects, to measure total global explanatory power;
- with country fixed effects, to focus on the within-country distribution of activity.

Interpretation. The coefficient vector tells us how lights vary across locations with different first-nature features.

3.3 Why country fixed effects matter

With country fixed effects, the regression becomes

$$\ln(\text{light}_{ic}) = X_{ic}\beta + f_c + \varepsilon_{ic}, \quad (3)$$

where f_c is a country fixed effect.

Interpretation.

- Country fixed effects absorb national institutions, national policy, and average income differences.
- The remaining identifying variation is across places *within* the same country.
- This helps separate direct geography effects from geography's indirect effect through national institutions.

3.4 Shapley decomposition and block-level explanatory power

To summarize the contribution of different groups of covariates, the paper computes Shapley values for:

- base variables,
- agriculture variables,
- trade variables,
- and country fixed effects.

Interpretation. The Shapley decomposition answers not only whether coefficients are individually significant, but also which groups of variables matter most for overall fit.

3.5 Heterogeneous geography by early and late development

The main heterogeneous-coefficient specification is

$$\ln(\text{light}_{ic}) = X_{ic}\beta + \text{Early}_c X_{ic}\beta_d + f_c + \varepsilon_{ic}. \quad (4)$$

Here, Early_c is an indicator for whether country c is an early agglomerator according to 1950 schooling, urbanization, or GDP per capita.

Interpretation. This equation asks whether the same geography variable has a different relationship with economic density in countries that urbanized early versus late.

3.6 Data-driven sample split

Rather than choose the early/late cutoff arbitrarily, the paper uses a Durlauf-Johnson style procedure:

- for each possible cutoff in the country ranking, split the sample into two groups;
- estimate the same geography regression separately in both groups;
- choose the cutoff that minimizes the summed residual sum of squares.

Interpretation. The split is chosen to maximize how much geography coefficients truly differ across the two groups.

3.7 Constrained differential-coefficient model

To summarize the group differences more parsimoniously, the paper estimates

$$\ln(\text{light}_{ic}) = X_{ic}^B \beta^B + X_{ic}^A \beta^A + X_{ic}^T \beta^T + \text{Early}_c [\alpha X_{ic}^A \beta^A + \gamma X_{ic}^T \beta^T] + f_c + \varepsilon_{ic}. \quad (5)$$

Here:

- X^B are base variables,
- X^A are agriculture variables,
- X^T are trade variables,
- α scales the agriculture block in early developers,
- γ scales the trade block in early developers.

Interpretation.

- $\alpha > 0$ means agriculture matters more in early developers.
- $\gamma < 0$ means trade matters less in early developers.

3.8 Standard errors and spatial correlation

The main fixed-effects regressions cluster errors in **3 by 3 sets of quarter-degree grid cells**. The paper also reports that Conley standard errors with a 40 km kernel are only about **5–20%** larger than the baseline clustered errors.

Interpretation. The results are not being driven by spurious precision from highly spatially correlated observations.

3.9 Spatial inequality regressions

At the country level, the paper relates spatial inequality to early-development measures using regressions of the form

$$\text{Gini}_c = a + b \text{EarlyMeasure}_{c,1950} + u_c, \quad (6)$$

and then adds controls for the geography-predicted Gini, land area, and population.

Interpretation. These regressions ask whether countries that agglomerated earlier are still more spatially equal today, over and above what raw geography alone would predict.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the baseline estimates

The baseline within-country results are identified by comparing places inside the same country that differ in geography. With country fixed effects, the paper is not asking whether Norway is richer than Niger; it is asking why some parts of a given country are much brighter than others.

Interpretation. This is the cleanest way to isolate direct geography effects from country-level institutions and policy.

4.2 Why lights are a sensible proxy for economic density

The authors compare subnational lights to population density and GDP per capita in outside data.

- Without country fixed effects, the R^2 of lights on population density alone is **0.530**, versus **0.285** for income per capita alone.
- With country demeaning, the corresponding R^2 values are **0.775** for population density and only **0.128** for income per capita.

Interpretation. Within countries, lights mostly trace where people live and work, not just where productivity per person is somewhat higher.

4.3 Why cross-section can still speak to history

A natural objection is that the data are essentially cross-sectional. The paper's answer is persistence. If city locations are durable, then the modern spatial distribution preserves information about the geography that mattered when urbanization began.

Interpretation. The paper is not pretending to observe the past directly. It infers the past from the durable imprint of past agglomeration.

4.4 Why the early-late result is not just Old World versus New World

The authors split the sample separately within the Old World and the New World. The agriculture-versus-trade differential remains present in both regions.

Interpretation. The headline result is not merely a colonization-reset story, though the paper notes that the New World often gives an even cleaner experiment.

4.5 Why variance differences are not the whole story

The pattern is not mechanically driven by one set of variables having much more within-country variance in one sample than the other.

- All five trade variables actually have larger variance in the early agglomerators.
- Eight of the 17 agriculture variables have larger variance in the late agglomerators.

Interpretation. The key result comes from different *weights* on geography, not merely different dispersion of geography.

4.6 How the paper handles spillovers and spatial dependence

The authors openly note that neighboring locations can affect each other. They respond in three ways:

- using quarter-degree cells rather than tiny pixels,
- clustering standard errors spatially,
- interpreting the coefficients as reduced-form geography effects that already embed agglomeration spillovers.

Interpretation. The paper does not claim to fully separate raw geography from all equilibrium spillovers. It instead estimates the combined reduced-form mapping from geography to activity.

4.7 What the paper can and cannot distinguish

- It can distinguish strong agriculture versus trade heterogeneity across early and late developers.
- It can show that spatial inequality is lower in early developers.
- It cannot cleanly assign all of the effect to only agricultural productivity and transport-cost timing, because democratization, resource extraction, and changing urban technologies may also matter.
- It cannot separately identify persistence from every alternative source of long-run path dependence.

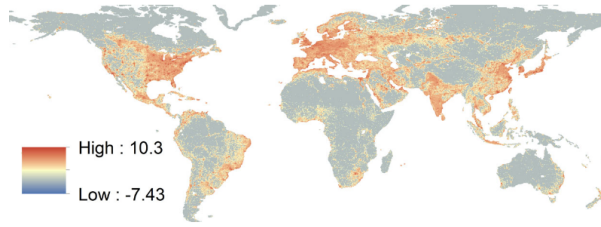


FIGURE I
Demeaned $\ln(\text{lights})$
Full-color figure is available in the online version of this article.

Interpretation. The paper is strongest on a coherent historical pattern, not on a single sharp instrument.

4.8 Relation to the literature and policy relevance

The paper contributes to the geography literature by:

- showing that first-nature geography still has large predictive power globally;
- connecting first nature to persistence and timing of development;
- shifting attention from cross-country levels to within-country spatial distributions.

It also speaks to policy debates on developing hinterland cities in places such as China and Sub-Saharan Africa.

Interpretation. The spatial pattern of rich countries may partly reflect old technology rather than a blueprint that late developers should copy today.

5 Tables and figures

5.1 Figure 1 and Table 1: the global lights map and baseline coefficients

Read:

- Figure 1 maps demeaned log lights across the world and shows dense clusters in Europe, eastern North America, India, coastal China, and parts of Latin America.
- Table 1 reports summary statistics and the baseline coefficients with and without country fixed effects.
- The full regression R^2 is **0.467** without country fixed effects and **0.577** with them.
- In the fixed-effects regression, land suitability has a coefficient of about **2.226**; harbor within 25 km has about **1.260**; coast has about **0.199**; distance to coast has about **-0.656**; growing days has about **0.0085**; malaria has about **-0.0472**.

Economic message. Geography has strong predictive power even in a simple linear specification, and the qualitative signs are sensible: better farm land and better transport access are associated with more activity.

TABLE I
SUMMARY STATISTICS AND BASELINE REGRESSION RESULTS

	Summary statistics		Regression w/out FEs		Regression w/ FEs	
	Mean (std. dev.)	Min, max	Coefficient	Shapley	Coefficient	Shapley
	(1)	(2)	(3)	(4)	(5)	(6)
Dependent variable ln(light/land pixels)	-3.357 (3.119)	-5.684 6.941				
Base covariates						
Ruggedness (OOOs)	2.781 (4.852)	0 95.81	-0.00764*** (0.00196)	0.000505 (0.00165)	-0.0148*** (0.00165)	0.000935 (0.00165)
Malaria index	1.921 (5.289)	0 38.08	-0.0340*** (0.00248)	0.0181 (0.00235)	-0.0472*** (0.00235)	0.0129 (0.00235)
Agriculture covariates						
Tropical moist forest	0.117 (0.321)	0 1	-0.0126 (0.0750)	0.165 (0.0651)	-0.207*** (0.0651)	0.130 (0.0651)
Tropical dry forest	0.0223 (0.148)	0 1	0.995*** (0.0942)	0.244*** (0.0796)	0.244*** (0.0796)	0.244*** (0.0796)
Temperate broadleaf	0.104 (0.306)	0 1	1.795*** (0.0701)	1.304*** (0.0647)	1.304*** (0.0647)	1.304*** (0.0647)
Temperate conifer	0.0330 (0.179)	0 1	0.776*** (0.0815)	0.161** (0.0777)	0.161** (0.0777)	0.161** (0.0777)
Boreal forest	0.166 (0.372)	0 1	-0.483*** (0.0758)	-1.283*** (0.0808)	-1.283*** (0.0808)	-1.283*** (0.0808)
Tropical grassland	0.121 (0.326)	0 1	-0.803*** (0.0555)	-0.0349 (0.0479)	-0.0349 (0.0479)	-0.0349 (0.0479)
Temperate grassland	0.0772 (0.267)	0 1	0.744*** (0.0649)	0.938*** (0.0571)	0.938*** (0.0571)	0.938*** (0.0571)
Montane grassland	0.0334 (0.180)	0 1	0.613*** (0.0798)	0.719*** (0.0716)	0.719*** (0.0716)	0.719*** (0.0716)
Tundra	0.122 (0.327)	0 1	-0.846*** (0.0848)	-1.417*** (0.0885)	-1.417*** (0.0885)	-1.417*** (0.0885)
Mediterranean forest	0.0242 (0.154)	0 1	0.843*** (0.0926)	1.362*** (0.0885)	1.362*** (0.0885)	1.362*** (0.0885)
Mangroves	0.00404 (0.0634)	0 1	0.0228 (0.160)	-0.443*** (0.138)	-0.443*** (0.138)	-0.443*** (0.138)
Desert	0.175 (0.380)	0 1				
Temperature (deg. C)	10.02 (13.77)	-22.29 30.37	0.172*** (0.00335)	0.0383 (0.00378)	0.116*** (0.00378)	0.0295 (0.00378)
Precipitation (mm/month)	60.82 (59.27)	0.387 921.9	-0.00897*** (0.000404)	0.0112 (0.000413)	-0.0113*** (0.000413)	0.0102 (0.000413)
Growing days	139.6 (99.04)	0 366	0.00989*** (0.000276)	0.0446 (0.000275)	0.00851*** (0.000275)	0.0364 (0.000275)
Land suitability	0.275 (0.320)	0 1	2.692*** (0.0545)	0.125 (0.0521)	2.226*** (0.0521)	0.102 (0.0521)
Abs(latitude)	38.31 (20.93)	0.125 74.88	0.114*** (0.00247)	0.0268 (0.00328)	0.0338*** (0.00328)	0.0144 (0.00328)
Elevation (km)	0.605 (0.790)	-0.187 6.169	0.521*** (0.0239)	0.00640 (0.0255)	0.0727*** (0.0255)	0.00536 (0.0255)

TABLE I
(CONTINUED)

	Summary statistics		Regression w/out FEs		Regression w/ FEs	
	Mean (std. dev.)	Min, max	Coefficient	Shapley	Coefficient	Shapley
	(1)	(2)	(3)	(4)	(5)	(6)
Trade covariates	0.0972 (0.296)	0 1	0.191*** (0.0373)	0.00254 (0.0300)	0.199*** (0.0300)	0.00222 (0.0300)
Coast	0.486 (0.481)	0 2.274	-0.685*** (0.0275)	0.0102 (0.0275)	-0.656*** (0.0318)	0.00770 (0.0318)
Distance to coast (000s km)	0.0273 (0.163)	0 1	1.456*** (0.0652)	0.0148 (0.0652)	1.260*** (0.0546)	0.0119 (0.0546)
Harbor < 25 km	0.0273 (0.163)	0 1	0.797*** (0.0623)	0.00246 (0.0623)	0.697*** (0.0569)	0.00213 (0.0569)
River < 25 km	0.0108 (0.104)	0 1	0.614*** (0.0867)	0.000406 (0.0867)	0.598*** (0.0828)	0.000453 (0.0828)
Lake < 25 km						
Number of observations	242,184		242,184		242,184	
R ²			0.467		0.577	

Notes. The first two columns show means and standard deviations, and minima and maxima, for all geographic variables for the full sample. The third and fifth columns report OLS coefficient estimates from equation (1) on the full sample, with and without country fixed effects, respectively. Standard errors, clustered by 3 × 3 sets of grid squares, are in parentheses. *p < .1, **p < .05, ***p < .01. Columns (4) and (6) report the corresponding Shapley values for biomes as a group, and for all other right-hand-side variables individually. See text for variable definitions.

5.2 Figure 2 and Table 2: fitted geography and the relative importance of blocks

Read:

- Figure 2 compares fitted lights from the no-fixed-effects regression to fitted lights from the fixed-effects regression with the fixed effects suppressed.
- The correlation between the two fitted maps is **0.861**.
- Table 2 shows that geography variables alone explain **46.7%** of the global variation in lights.
- With country fixed effects, base plus agriculture variables lift the R^2 to **0.566**, while base plus trade variables only lift it to **0.370**.
- The corresponding Shapley values in the fixed-effects specification are **0.009** for base, **0.321** for agriculture, **0.025** for trade, and **0.222** for country fixed effects.

Economic message. Agriculture-related geography dominates trade-related geography in the average within-country decomposition, but that average masks huge heterogeneity across countries.

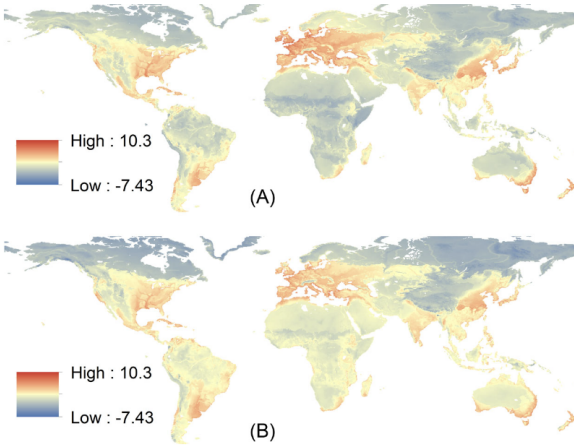


FIGURE II

Panel A: Demeaned Predicted $\ln(\text{lights})$ without Fixed Effects. Panel B: Demeaned Predicted $\ln(\text{lights})$, Fixed Effects Specification with Fixed Effects Suppressed

Each map reports demeaned predicted values from a regression of $\ln(\text{lights})$ on all geographic variables. In Panel B, the regression is run with country fixed effects, but predicted values are calculated setting those fixed effects to zero. Full-color figure is available in the online version of this article.

TABLE II
 R^2 AND SHAPLEY VALUES FROM REGRESSIONS PREDICTING $\ln(\text{LIGHT/LAND PIXELS})$

	No country FEs (1)	With country FEs (2)
Panel A: R^2		
All variables ($N = 242,184$)	0.467	0.577
Base variables (malaria, ruggedness)	0.020	0.355
Agriculture variables (plus base)	0.450	0.566
Trade variables (plus base)	0.066	0.370
Country fixed effects		0.345
Panel B: Shapley values		
Base	0.011	0.009
Agriculture	0.423	0.321
Trade	0.033	0.025
Country FEs		0.222

Notes. Each entry in Panel A represents an R^2 value from a separate regression of $\ln(\text{light})$ on the right-hand-side variables listed in the row and column headings. Each column in Panel B corresponds to a separate regression. The values shown are Shapley values for the set of variables shown.

5.3 Figure 3 and Figure 4: preliminary evidence and the timing story

Read:

- Figure 3 plots each country's coastal-versus-interior residual differential against average years of schooling in 1950.
- The slope is about **-0.342** with standard error **0.065** and $R^2 = 0.20$.
- Low-schooling countries have especially large positive coastal residuals after removing average geography effects.
- Figure 4 plots literacy paths for early- and late-agglomerator countries against the global freight index. Early developers had relatively high literacy before the big late-nineteenth-century drop in freight costs; late developers did not.

Economic message. The raw residual patterns and the historical timing data line up with the paper's mechanism: transport access matters especially strongly where urbanization began only after transport had already become cheap.

5.4 Figure 5 and Table 3: data-driven cutoffs and differential explanatory power

Read:

- Figure 5 chooses the early/late split by minimizing summed residual variance. The optimal cutoffs are **2.83** years of schooling, **36.16%** urbanization, and **\$2,231** GDP per capita in 1950.

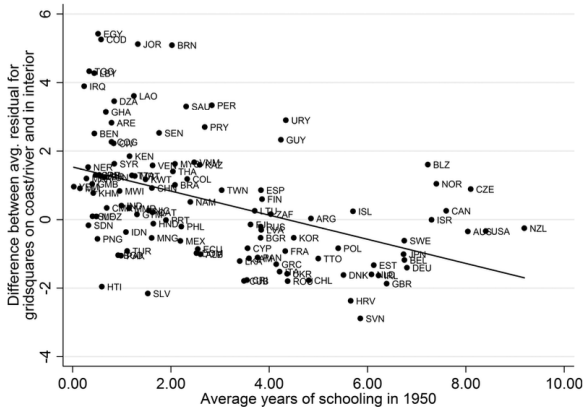


FIGURE III

Difference between Average Coastal/River and Interior Residuals by Years of Schooling in 1950

In(lights) are first regressed on all geographic variables in the global sample with country fixed effects, and residuals are calculated suppressing the fixed effects. These residuals are averaged separately within each country for two groups: cells within 25 km of a coast or navigable river and those farther away. The difference between these averages is the height of each point.

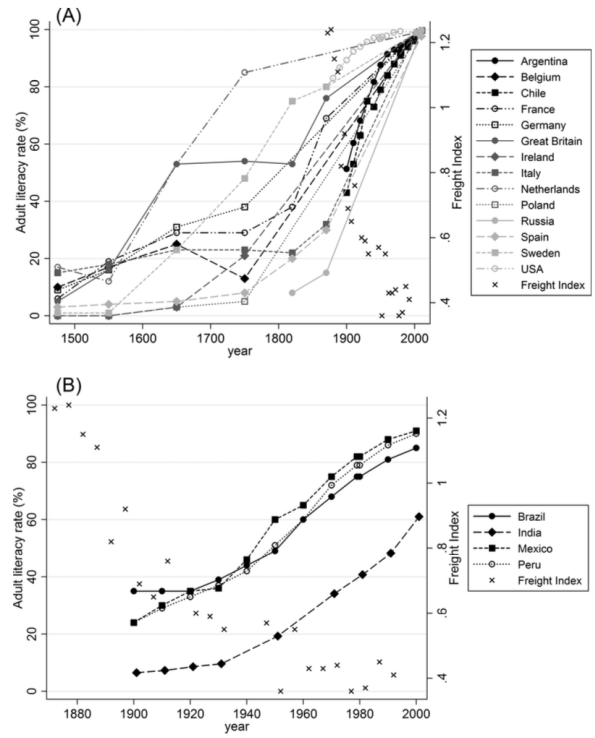


FIGURE IV

Panel A: Global Transport Costs and High Education Country Literacy Rates
Panel B: Global Transport Costs and Low Education Country Literacy Rates

The global real freight index is from [Mohammed and Williamson \(2004\)](#). Periods including world war years are omitted. Literacy rates for all countries except India are from [Roser and Ortiz-Ospina \(2016\)](#). Literacy rates for India are from [UNESCO \(1957\)](#), [Ministry of Human Resource Development \(1987\)](#), and [World Bank \(2015\)](#).

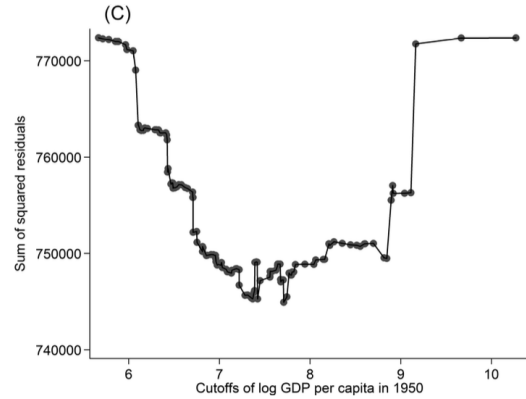
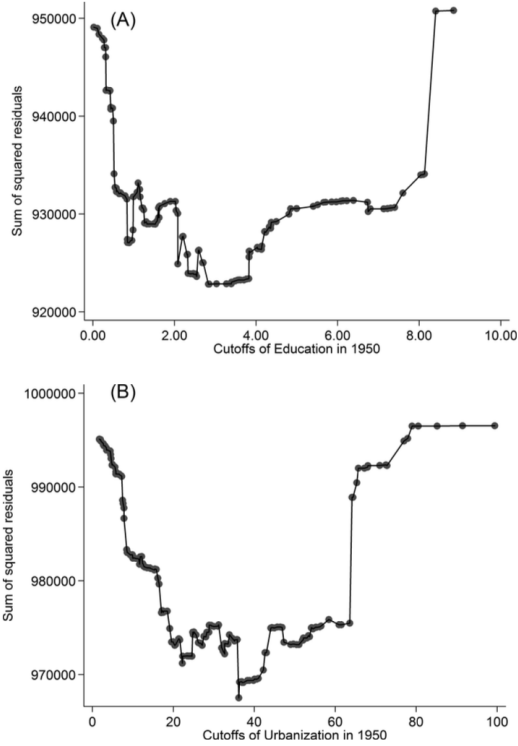


FIGURE V
(CONTINUED)

Panel A: Years of Schooling in 1950: Total SSR
 Panel B: Urbanization in 1950: Total SSR
 Panel C: GDP per Capita in 1950: Total SSR

In each panel, the vertical coordinate of each point represents the sum of squared residuals summed across two regressions on two disjoint samples, one each for countries above and below the cutoff of the cut variable specified on the horizontal axis. Each regresses $\ln(\text{light})$ on all geographic variables and country fixed effects. Each point corresponds to an individual country in the sample (i.e., the exercise is run for each country's value of the cut variable and ranked by the cut variable).

TABLE III
 R^2 DIFFERENTIALS OF TRADE AND AGRICULTURE VARIABLES IN REGRESSION
PREDICTING LN (LIGHT/LAND PIXELS) FOR HIGH/LOW EDUCATION AND URBANIZATION
COUNTRIES

	Education		Urbanization		GDP per capita	
	High	Low	High	Low	High	Low
Countries	58	82	63	121	36	101
Observations	126,671	100,361	138,020	103,975	80,310	100,602
Panel A: R^2						
Full sample						
Base + FE	0.385	0.294	0.351	0.362	0.387	0.375
Agriculture + base + FE	0.653	0.452	0.614	0.511	0.644	0.521
Trade + base + FE	0.425	0.395	0.386	0.452	0.419	0.467
High - Low double differential	0.171		0.170		0.171	
Panel B: Shapley values						
Full sample						
Base	0.006	0.020	0.005	0.022	0.004	0.021
Agriculture	0.397	0.197	0.371	0.217	0.358	0.216
Trade	0.029	0.091	0.026	0.080	0.032	0.093
Country FEs	0.227	0.182	0.218	0.223	0.255	0.224
Panel C: R^2 , hemispheres						
New World						
Base + FE	0.245	0.236	0.253	0.258	0.239	0.264
Agriculture + base + FE	0.609	0.346	0.586	0.394	0.581	0.399
Trade + base + FE	0.303	0.321	0.297	0.343	0.286	0.348
High - low double differential	0.280		0.238		0.244	
Old World						
Base + FE	0.486	0.345	0.436	0.409	0.420	0.425
Agriculture + base + FE	0.706	0.528	0.661	0.569	0.611	0.580
Trade + base + FE	0.518	0.433	0.467	0.485	0.450	0.504
High - low double differential	0.092		0.111		0.085	

Notes. Each number in the first three rows of Panel A is an R^2 value from a separate regression of ln(light) on the set of right-hand-side variables listed in the row, for a sample defined by the column headings. The last row shows the double differential (Agriculture high – Trade high) – (Agriculture low – Trade low). FE stands for country fixed effects. Panel B shows the corresponding Shapley values, and Panel C is the analog of Panel A run separately for the Old and New Worlds. The cutoffs for education, urbanization, and GDP per capita are, respectively: 2.83 years of schooling, 36.16% urbanized, and \$2,231 (2005 PPP).

- Table 3 shows that, for the education split, base plus country fixed effects produce $R^2 = 0.385$ in high-education countries and 0.294 in low-education countries.
- Adding agriculture raises the R^2 to **0.653** in high-education countries but only to **0.452** in low-education countries.
- Adding trade instead raises the R^2 only to **0.425** in high-education countries but to **0.395** in low-education countries.
- The implied double differential is about **0.171** for all three split variables.

Economic message. Geography matters differently depending on when a country developed. Agriculture dominates in early developers; trade matters relatively more in late developers.

5.5 Table 4, Table 5, and Figure 6: marginal effects and the difference map

Read:

- Table 4 shows that trade variables have large positive effects in late developers and strongly offsetting interactions in early developers.
- For the education split, the main effect of being coastal is about **0.915** and the interaction for early developers is about **-0.995**; the main effect of distance to coast is about **-1.430** and the interaction is about **1.540**.
- Similar offsetting interactions appear for rivers, lakes, and harbors.

TABLE IV
REGRESSION RESULTS ALLOWING INTERACTIONS BETWEEN GEOGRAPHIC VARIABLES
AND EARLY AGGLOMERATOR DUMMY

	Education		Urbanization		GDP per capita	
	Main effect (1)	Interaction (2)	Main effect (3)	Interaction (4)	Main effect (5)	Interaction (6)
Base covariates						
Ruggedness (OOOs)	-0.0169*** (0.00283)	0.00995*** (0.00361)	-0.0189*** (0.00255)	0.00550* (0.00334)	-0.0154*** (0.00258)	0.0166*** (0.00388)
Malaria index	-0.0267*** (0.00273)	-0.0634*** (0.00863)	-0.0252*** (0.00245)	-0.124*** (0.00984)	-0.0247*** (0.00251)	-0.122*** (0.00936)
Agriculture covariates						
Tropical moist forest	-0.0667 (0.0802)	0.812*** (0.239)	0.259*** (0.0759)	-1.992*** (0.193)	0.290*** (0.0759)	-2.164*** (0.188)
Tropical dry forest	0.376*** (0.0912)	0.363 (0.336)	0.501*** (0.0910)	-0.540*** (0.190)	0.550*** (0.0907)	-0.722*** (0.191)
Temperate broadleaf	0.982*** (0.0912)	0.259* (0.137)	1.035*** (0.0870)	0.373*** (0.128)	0.994*** (0.0876)	0.145 (0.143)
Temperate conifer	0.322*** (0.120)	-0.170 (0.164)	0.573*** (0.116)	-0.419*** (0.156)	0.434*** (0.115)	-0.434*** (0.171)
Boreal forest	-0.0237 (0.133)	-1.041*** (0.172)	-0.0352 (0.136)	-0.882*** (0.169)	-0.223 (0.138)	-1.741*** (0.193)
Tropical grassland	-0.0181 (0.0659)	-0.360*** (0.107)	0.154*** (0.0587)	-0.428*** (0.101)	0.170*** (0.0589)	-0.688*** (0.105)
Temperate grassland	0.440*** (0.0880)	0.375*** (0.126)	0.370*** (0.101)	0.622*** (0.126)	0.0608 (0.111)	0.771*** (0.147)
Montane grassland	0.306*** (0.0988)	1.154*** (0.166)	0.462*** (0.0921)	0.947*** (0.152)	0.480*** (0.0928)	0.748*** (0.168)
Tundra	-0.717*** (0.116)		-0.634*** (0.107)		-1.585*** (0.148)	
Mediterranean forest	1.999*** (0.130)	-1.189*** (0.166)	1.951*** (0.130)	-1.083*** (0.165)	1.840*** (0.124)	-1.100*** (0.165)
Mangroves	-0.945*** (0.165)	0.765* (0.443)	-0.378** (0.164)	-1.116*** (0.323)	-0.442*** (0.163)	-0.909** (0.390)
Temperature (deg. C)	0.121*** (0.00790)	-0.0238*** (0.00903)	0.133*** (0.00768)	-0.0397*** (0.00881)	0.122*** (0.00774)	-0.148*** (0.0112)
Precipitation (mm/month)	-0.0101*** (0.000563)	0.00176** (0.000780)	-0.00943*** (0.000542)	0.000307 (0.000745)	-0.0103*** (0.000567)	0.0000965 (0.000816)
Growing days	0.00711*** (0.000398)	0.00129** (0.000551)	0.00716*** (0.000372)	0.00125** (0.000527)	0.00733*** (0.000377)	0.00166*** (0.000569)
Land suitability	2.158*** (0.0774)	-0.0960 (0.110)	2.070*** (0.0746)	0.0743 (0.104)	1.981*** (0.0755)	-0.139 (0.118)
Abs (latitude)	0.0886*** (0.00584)	-0.0921*** (0.00685)	0.102*** (0.00540)	-0.110*** (0.00661)	0.1000*** (0.00542)	-0.189*** (0.00790)
Elevation (km)	0.248*** (0.0408)	-0.635*** (0.0576)	0.222*** (0.0392)	-0.363*** (0.0543)	0.167*** (0.0395)	-0.711*** (0.0593)
Trade covariates						
Coast	0.915*** (0.0701)	-0.995*** (0.0767)	0.706*** (0.0640)	-0.669*** (0.0712)	0.735*** (0.0624)	-0.559*** (0.0747)
Distance to coast (OOOs km)	-1.430*** (0.0472)	1.540*** (0.0647)	-1.460*** (0.0469)	1.389*** (0.0632)	-1.512*** (0.0471)	1.367*** (0.0956)
Harbor < 25 km	1.564*** (0.104)	-0.345*** (0.123)	1.365*** (0.0949)	-0.129 (0.115)	1.216*** (0.0871)	0.0546 (0.115)
River < 25 km	1.208*** (0.104)	-0.772*** (0.120)	0.914*** (0.105)	-0.359*** (0.120)	0.944*** (0.106)	-0.520*** (0.130)
Lake < 25 km	0.762*** (0.133)	-0.346** (0.170)	0.548*** (0.149)	0.0182 (0.177)	0.730*** (0.137)	-0.310* (0.170)
N	227,032		241,995		180,912	

Notes. Each set of two consecutive columns reports OLS coefficient estimates from a separate regression of equation (2) on a global sample, split by education, urbanization, and GDP per capita, respectively, in 1950. The first column in each pair shows main terms, and the second column shows interaction terms. Standard errors, clustered by 3×3 sets of grid squares, are in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$.

TABLE V
DIFFERENTIAL COEFFICIENT RESULTS

	Education	Urbanization	GDP per capita
Agriculture differential (α)	0.332*** (0.0238)	0.193*** (0.0209)	0.254*** (0.0239)
Trade differential (γ)	-0.650*** (0.0178)	-0.393*** (0.0218)	-0.526*** (0.0321)
N	227,032	241,995	180,912

Notes. Each column reports nonlinear least squares estimates of α and γ in equation (3), for education, urbanization, and GDPpc split variables. Standard errors, clustered by 3×3 sets of grid squares, are in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$.

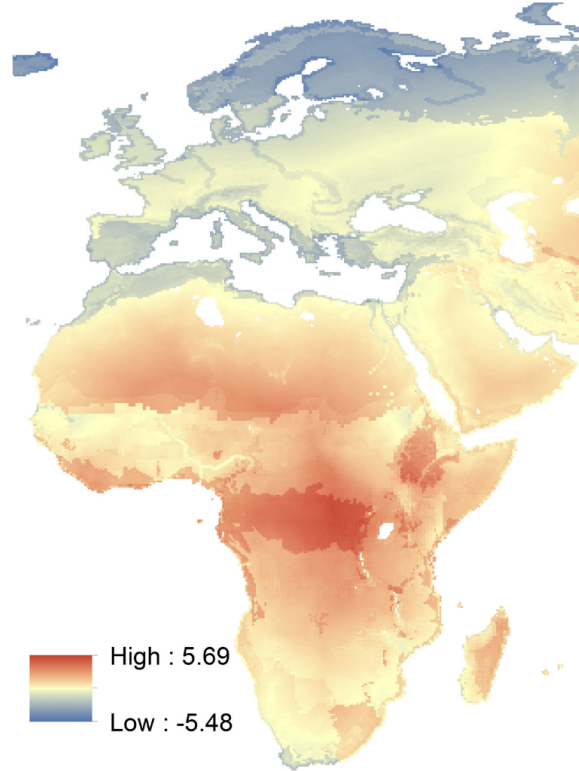


FIGURE VI
Demeaned Difference between High and Low Predicted Lights with Fixed Effects Suppressed
This map shows fitted values of $(\alpha X_{ic}^A \beta_A + \gamma X_{ic}^T \beta_T)$ from [equation \(3\)](#). Color figure available in online version of this article.

- Table 5 summarizes the group differences: for the education split, $\alpha = 0.332$ and $\gamma = -0.650$; for urbanization, $\alpha = 0.193$ and $\gamma = -0.393$; for GDP per capita, $\alpha = 0.254$ and $\gamma = -0.526$.
- Figure 6 maps fitted values of the difference between early-developer and late-developer geography weights. Interior areas of Africa would be denser under early-developer coefficients; many coastal areas would be denser under late-developer coefficients.

Economic message. The trade-versus-agriculture reversal is not only about overall fit. It also appears directly in the slope coefficients, especially for coastal-access variables.

5.6 Figure 7, Table 6, Figure 8, and Table 7: spatial inequality in lights and education

Read:

- Figure 7 shows a downward relationship between the spatial Gini of lights and years of schooling in 1950.
- In Table 6, the schooling coefficient is about **-0.0265** without controls and **-0.0263** with land area and population controls.

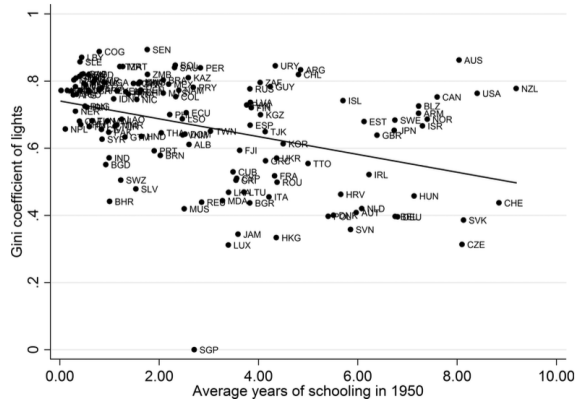


FIGURE VII
Gini Coefficient of Lights by Years of Schooling in 1950

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Education 1950	-0.0262*** (0.00569)	-0.0326*** (0.00554)	-0.0265*** (0.00445)						
Urbanization 1950				-0.00283*** (0.000768)	-0.00303*** (0.000648)	-0.00179*** (0.000495)			
ln(GDP per cap. exp) 1950							-0.0394*** (0.0145)	-0.0418*** (0.0189)	-0.0267 (0.0167)
Gini of predicted lights		0.249*** (0.0801)	0.0809 (0.0525)		0.315*** (0.0753)	0.0503 (0.0556)		0.189* (0.0995)	0.0284 (0.0684)
ln(land area)			0.0673*** (0.00988)		0.0704*** (0.00579)			0.0714*** (0.00690)	
ln(population in 2010)			-0.4886*** (0.00783)		-0.0433*** (0.00673)			-0.0512*** (0.00850)	
Constant	0.741*** (0.0163)	0.648*** (0.0391)	0.344*** (0.0672)	0.734*** (0.0212)	0.609*** (0.0331)	0.167*** (0.0586)	0.895*** (0.101)	0.896*** (0.112)	0.0286 (0.126)
<i>N</i>	140	140	139	184	184	181	137	137	135
<i>R</i> ²	0.157	0.242	0.600	0.126	0.241	0.583	0.033	0.077	0.511

Notes. Each column reports OLS coefficient estimates from a country-level regression of the Gini coefficient of lights on the variables shown. Robust standard errors are in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$.

- The paper notes that a one-standard-deviation increase in 1950 schooling (**2.35** years) lowers the lights Gini by about **0.06**, or about **0.40** standard deviations.
- Figure 8 shows that the population-weighted regional education Gini also declines with 1950 schooling.
- Table 7 reports a schooling coefficient of about **-0.0183** without controls and **-0.0193** with controls for land area and population.

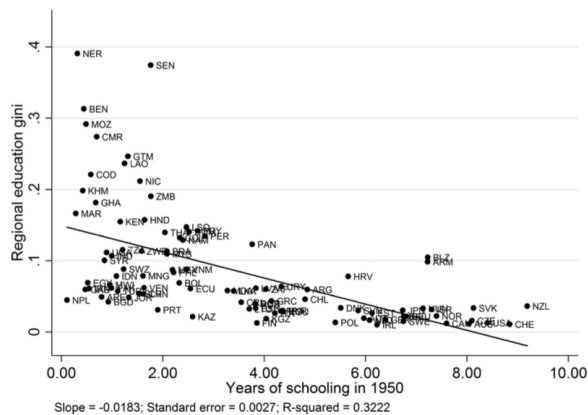


FIGURE VIII
Population-Weighted Regional Education Gini by Years of Schooling in 1950

	(1)	(2)	(3)	(4)	(5)	(6)
Years of schooling in 1950	-0.0183*** (0.00276)	-0.0193*** (0.00285)				
Urbanization in 1950			-0.00224*** (0.000362)	-0.00228*** (0.000368)		
Log GDP per capita in 1950					-0.0532*** (0.00939)	-0.0576*** (0.0103)
Log area (sq km)		0.00656 (0.00533)		0.0119** (0.00562)		0.0152** (0.00698)
Log population in 2010		-0.0131** (0.00575)		-0.0101* (0.00597)		-0.0207** (0.00819)
Constant	0.149*** (0.0149)	0.198*** (0.0474)	0.163*** (0.0175)	0.115** (0.0451)	0.496*** (0.0761)	0.542*** (0.103)
Observations	97	97	106	106	88	88
R ²	0.322	0.354	0.305	0.330	0.278	0.336

Notes. Each column reports OLS coefficient estimates from a country-level regression of the Gini coefficient of years of schooling on the variables shown. Robust standard errors are in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$.

Economic message. Early developers are not only more interiorly dispersed in economic activity; they are also more equal in the spatial distribution of human capital.

6 Short summary

- The paper explains the modern spatial distribution of economic activity with a compact set of geography variables plus history.

- Geography alone explains almost half of worldwide lights variation and about one-third of within-country variation.
- The central puzzle is that agricultural geography matters more in today's rich early developers, while trade geography matters relatively more in today's poorer late developers.
- The paper resolves the puzzle through persistence and timing: early developers urbanized before transport became cheap; late developers urbanized after.
- Trade-related variables, especially coastal access, have much stronger effects in late developers, while agriculture variables are relatively stronger in early developers.
- Countries that developed earlier are also more spatially equal in both lights and schooling.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that first-nature geography still explains a great deal of the modern location of economic activity, but the relevant geography is not weighted the same way everywhere. Countries that agglomerated early still bear the imprint of agricultural geography; countries that agglomerated late are more shaped by trade geography.

7.2 Economic intuition

When urbanization began under high transport costs, cities had to sit near good farming regions. Once transport costs later fell, those cities persisted. When transport was already cheap before large-scale urbanization began, manufacturing could instead concentrate in a few well-connected places, often on the coast. Persistence then locked in those spatial patterns.

7.3 Takeaway

The map of rich countries is not necessarily the efficient map for countries urbanizing today. The paper's message is that nature matters, but history determines which parts of nature continue to matter most.